

Ritajit Kundu

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Education and Experience

Postdoctoral Researcher, Indiana University Bloomington, USA	Sep 2026–Present
ANRF National Postdoctoral Fellow, The Institute of Mathematical Sciences, India	Jan 2026–Aug 2026
FARE Postdoctoral Fellow, Indian Institute of Technology Kanpur, India	Jan 2025–Dec 2025
Ph.D. in Physics, Indian Institute of Technology Kanpur, India	Jun 2025
M.Sc. in Physics, Indian Institute of Technology Kanpur, India	2019
B.Sc. in Physics with Honours, Calcutta University	2017

Research Interests

Quantum Condensed Matter Physics: Moiré materials, fractional quantum Hall effect, and network models.

Technical Skills

Theoretical Techniques

Many-Body Physics: Hartree–Fock theory and variational Monte Carlo.

Moiré Systems: Continuum models for moiré systems.

Quantum Transport: Kubo formula and nonequilibrium Green’s-function methods.

Programming Languages

Experienced with C++, Python, Mathematica, Julia, and Bash scripting.

Publications [Google Scholar](#), [arXiv](#)

(† Equal Contribution)

- [1] J. Sarkar, A. Mukherjee, A. Basu, **R. Kundu**, A. Kundu, M. Deshmukh, van der Waals Josephson junctions for quantum science and technology, [Nat. Rev. Phys.](#) (2026).
- [2] A. Basu, S. Ash, **R. Kundu**, N. Bhatia, S. Nema, T. Gawade, K. Agrawal, A. Shah, R. Kulkarni, D. Jangade, A. Kundu, A. Thamizhavel, M. Deshmukh, Quantum interference in a twisted high-Tc SQUID senses emergent interfacial order, [Nat. Commun.](#) **17**, 8310 (2026).
- [3] A. Mukherjee, S. Layek, S. Sinha, **R. Kundu**, A. H. Marchawala, M. Hingankar, J. Sarkar, L. D. Sangani, H. Agarwal, S. Ghosh, et al., Superconducting magic-angle twisted trilayer graphene hosts competing magnetic order and moiré inhomogeneities, [Nat. Mater.](#) **24**, 1400–1406 (2025).
- [4] A. Dubey†, **R. Kundu**†, A. Kundu, Time-resolved ARPES and optical transport properties of irradiated twisted bilayer graphene in a steady state, [Phys. Rev. B](#) **111**, 035431 (2025).
- [5] **R. Kundu**, A. Kundu, Josephson junction of minimally twisted bilayer graphene, [Phys. Rev. B](#) **110**, 085422 (2024).
- [6] **R. Kundu**, H. A. Fertig, A. Kundu, Broken symmetry and competing orders in Weyl semimetal interfaces, [Phys. Rev. B](#) **107**, L041402 (2023).

- [7] R. Mukherjee, **R. Kundu**, A. Singh, A. Kundu, Schwinger-boson mean-field study of the spin-1/2 J_1 - J_2 - J_χ model in the honeycomb lattice: Thermal Hall signature, [Phys. Rev. B **107**, 155122 \(2023\)](#).
- [8] S. Mohapatra, **R. Kundu**, A. Dubey, D. Dutta, A. Singh, Role of orbital off-diagonal spin and charge condensates in a three orbital model for Ca_2RuO_4 : Coulomb renormalized spin-orbit coupling, orbital moment, and tunable magnetic order, [J. Magn. Magn. Mater. **537**, 168172 \(2021\)](#).

Awards

National Postdoctoral Fellowship Anusandhan National Research Foundation	Jan 2026–Aug 2026
Fellowship for Academic and Research Excellence Indian Institute of Technology Kanpur	Jan 2025–Dec 2025
Prime Minister’s Research Fellow MHRD, Government of India Showcase of research work	2019–2024
Certificate of Merit Indian Institute of Technology Kanpur For academic excellence in the Master of Science program	2017

Selected Schools and Conferences

Synthetic Electronic Quantum Matter (SELEQ) Broken symmetry and competing orders in Weyl semimetal interfaces with Fermi arc states DIPC, Donostia-San Sebastian, Spain (Poster presentation in-person)	2024
Workshop on Twistronics and Moiré Materials: Bridging Theory and Experiments Josephson junction in minimally twisted bilayer graphene ICTP, Trieste, Italy (Poster presentation in-person)	2024
Condensed Matter meets Quantum Information Broken symmetry and competing orders in Weyl semimetal interfaces with Fermi arc states ICTS, Bengaluru, India (Poster presentation in-person)	2023
APS March Meeting SNS junction in small-angle twisted bilayer graphene (Contributed talk online)	2023
Annual Conference on Quantum Condensed Matter (QMAT) Broken symmetry and competing orders in Weyl semimetal interfaces with Fermi arc states IIT Kanpur, India (Poster presentation volunteer in-person)	2022
APS March Meeting Broken symmetry and competing orders in Weyl semimetal interfaces with Fermi arc states (Contributed talk online)	2022

Teaching Activities

Tutorials at NPTEL

- Advanced Condensed Matter Physics (Jan–Mar 2023), [YouTube playlist](#).
- Advanced Quantum Mechanics with Applications (Jul–Sep 2022), [YouTube playlist](#).

Kendriya Vidyalaya

Conducted weekly problem-solving sessions for Class XII students at Kendriya Vidyalaya, IIT Kanpur Campus (2021).

IIT Kanpur

- Advanced Statistical Mechanics, PHY613 (2024 odd semester at IITK | instructor: Adhip Agarwala).
- Undergraduate Lab, PHY101 (2021 odd semester at IITK).
- Quantum Many-Body Physics, PHY607 (2020 even semester at IITK | instructor: Arijit Kundu).
- Condensed Matter Physics, PHY543 (2020 odd semester at IITK | instructor: Arijit Kundu).
- Electronics, PHY441 (2019 odd semester at IITK | instructor: Jayita Nayak).

References

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